The Spiral of Theodorus

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pythagorean-spiral — a Typst package · v0.1.0

1. The construction

The **spiral of Theodorus** (also called the **snail of Pythagoras**, “escargot de Pythagore”) is built from right triangles sharing a common vertex O:

- triangle 1 is right-angled isosceles with legs of length a;
- every following triangle has one leg equal to the previous hypotenuse and the **other leg still equal to a** (the new leg added at each step always has the length a of the starting segment).

Hence, for any starting length a, the n-th triangle has legs $a\sqrt{n}$ and a, and hypotenuse $a\sqrt{n+1}$:

Triangle	**Legs**	**Hypotenuse**
1	a, a	$a\sqrt{2}$
2	$a\sqrt{2}$, a	$a\sqrt{3}$
3	$a\sqrt{3}$, a	$a\sqrt{4}$
$\vdots$	$\vdots$	$\vdots$
n	$a\sqrt{n}$, a	$a\sqrt{n+1}$

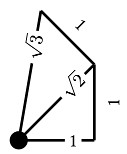
The angle added at step n is

$$\theta_n = \arctan\left(\frac{1}{\sqrt{n}}\right)$$

so after N triangles the spiral has turned by

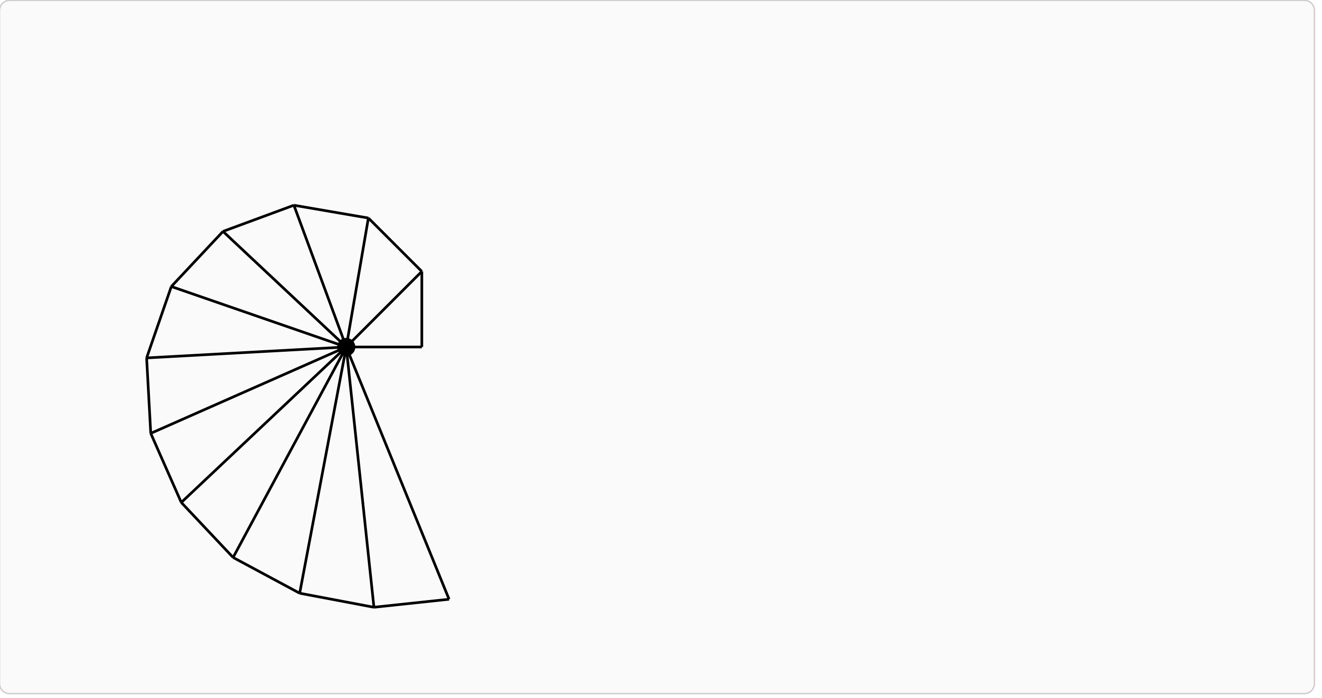
$$\varphi_N = \sum_{n=1}^{N} \arctan\left(\frac{1}{\sqrt{n}}\right)$$

which famously exceeds a full turn already at $N = 17$: $\varphi_{17} \approx 364.78°$.



The length labels are in units of a: each new leg has length a (shown as 1), the rays have length $a\sqrt{1}$, $a\sqrt{2}$, $a\sqrt{3}$, ….

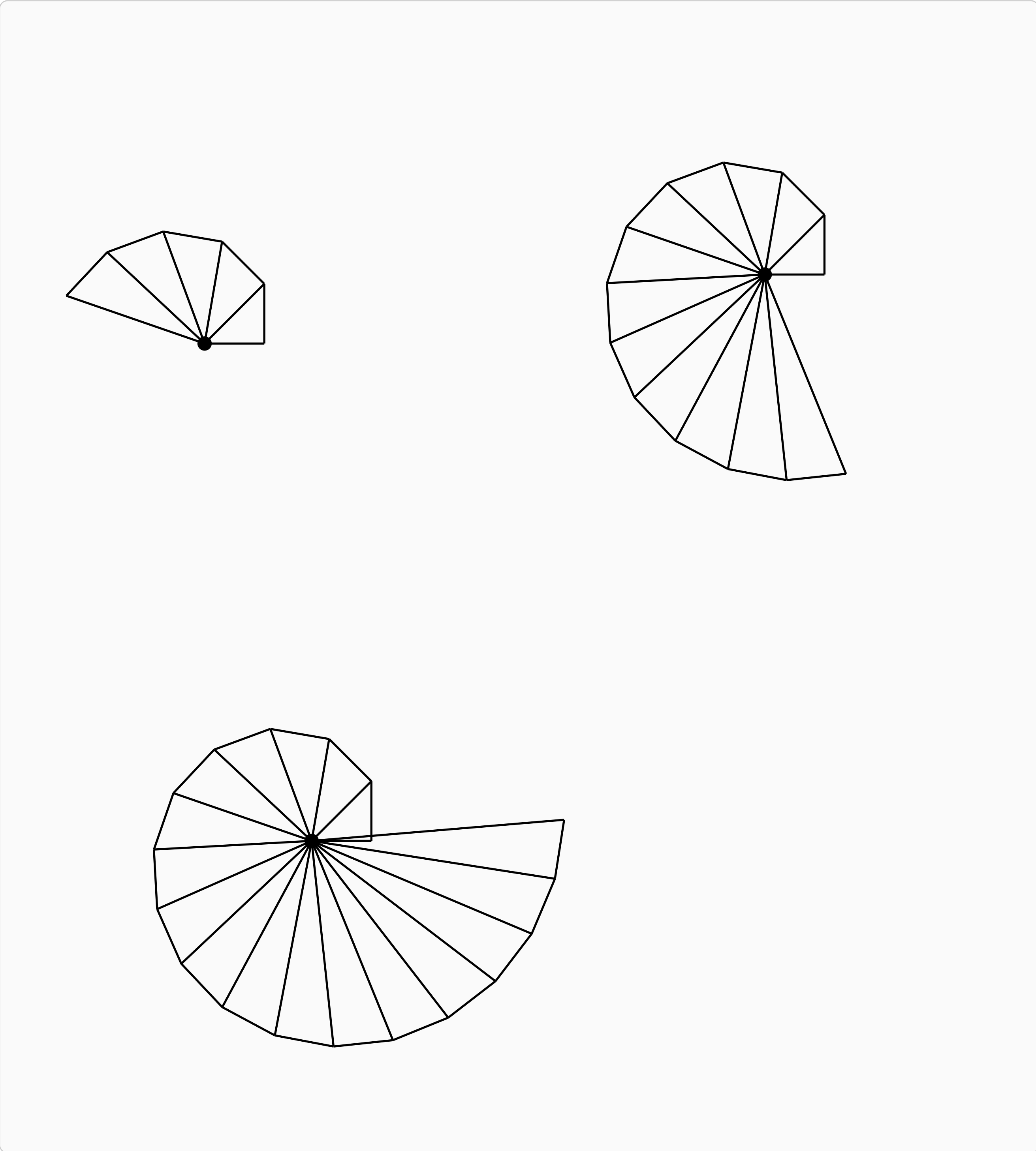
2. Basic usage



The package works with pure Typst primitives: the result is a normal box that measures, nests and flows like any other content.

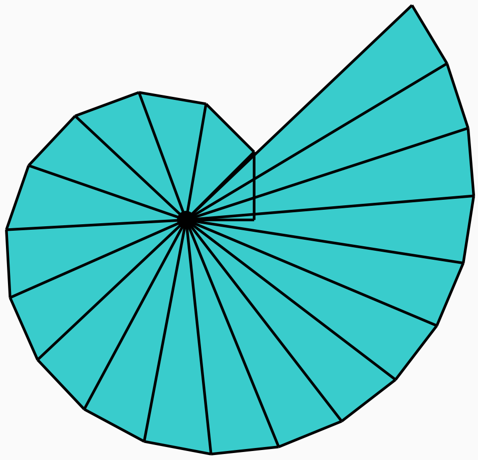
3. Parameters

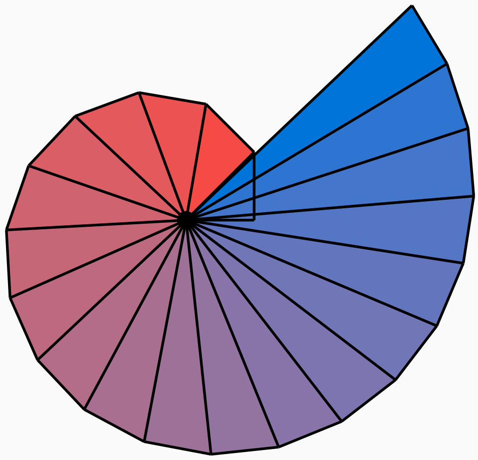
3.1. steps and size

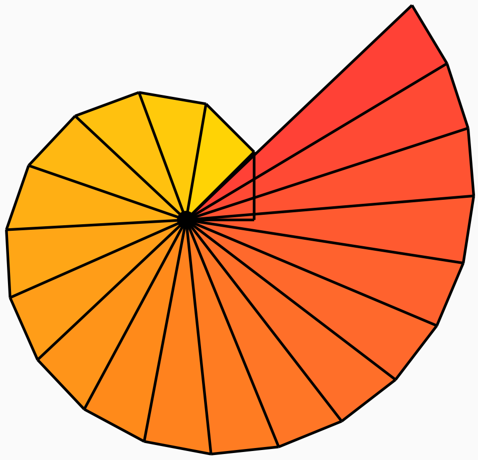


3.2. fill

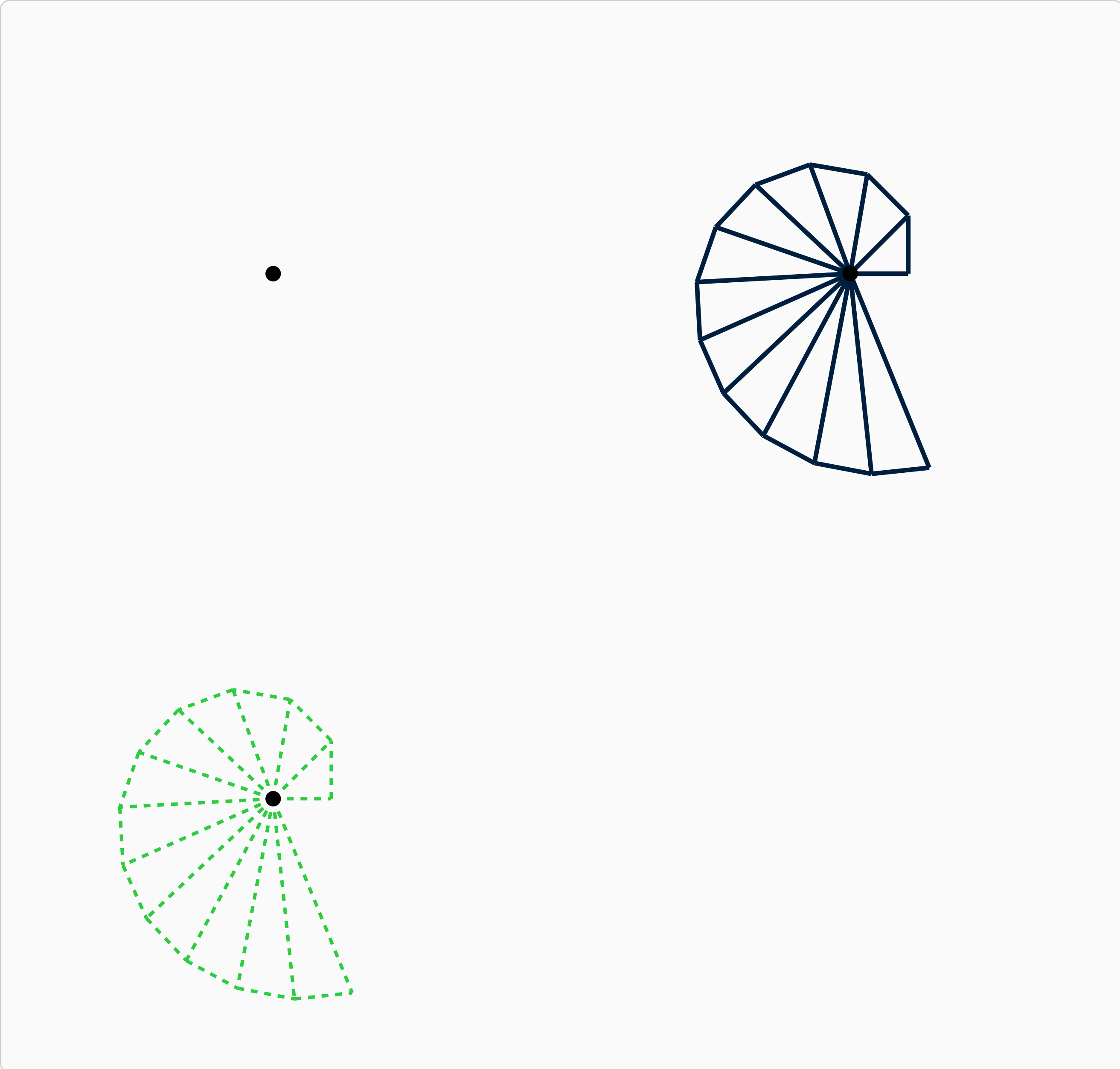
`fill` accepts a colour, a gradient (sampled along the spiral) or an array of colours (interpolated along the spiral):



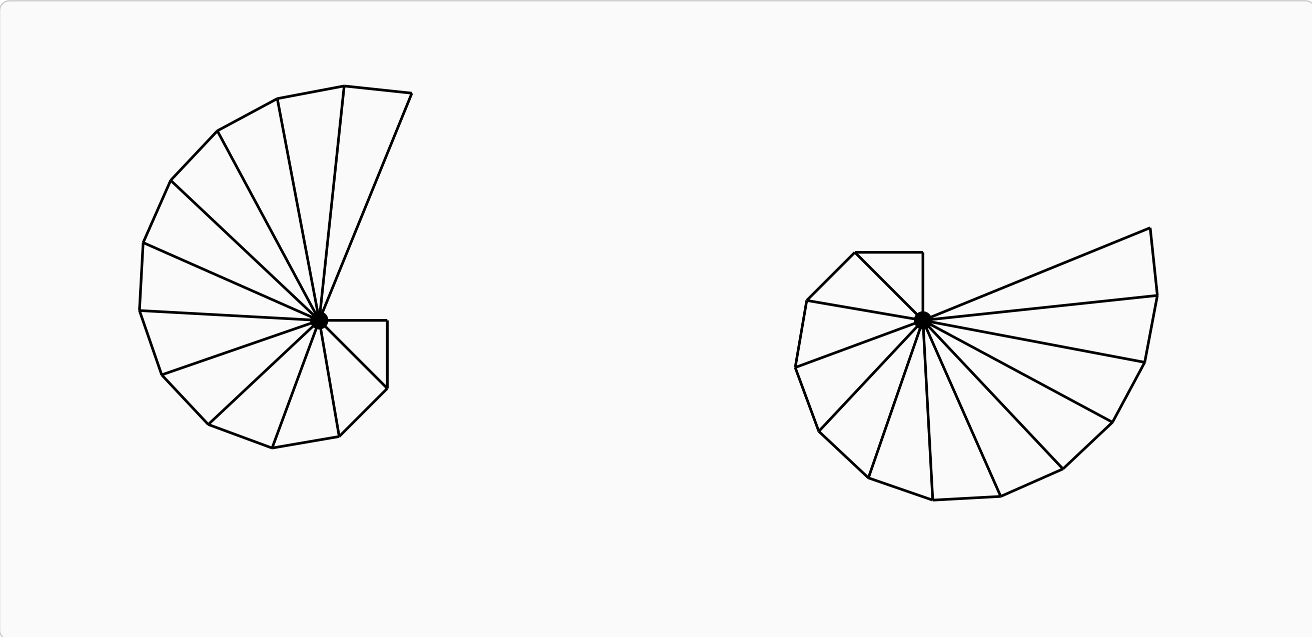




3.3. stroke

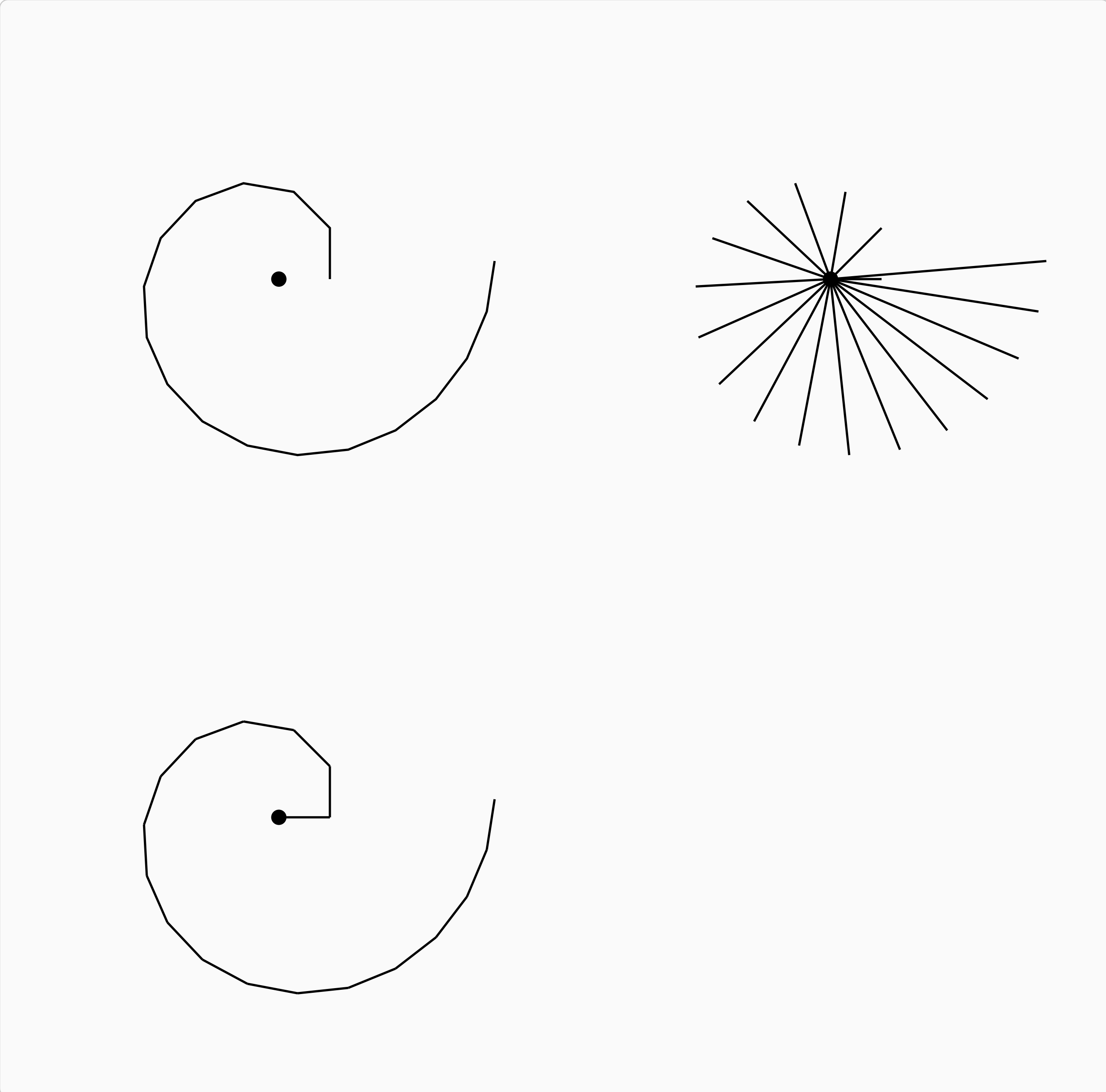


3.4. direction and start-angle



3.5. mode

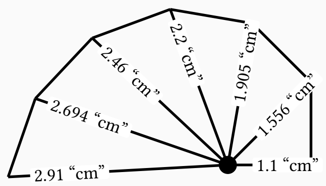
- "`triangles`" — full outlines (default);
- "`spiral`" — only the outer staircase $P_0 \to P_1 \to \ldots \to P_N$;
- "`rays`" — only the segments from the centre.

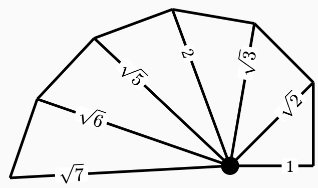


3.6. Length labels

Write the exact length of every ray. The ray closing the k-th triangle ($O \to P_{k-1}$) has length $a\sqrt{k}$, where a is the length of the starting leg and k is the step number — the starting leg itself is step 1 ($a\sqrt{1} = a$). Each label is **rotated parallel to its ray** and stays on it, positioned along the ray by `length-pos` (default 0.72: 0 = at the centre O, 1 = at the outer endpoint P_n) — near the outer end the rays are further apart, so the labels do not overlap. `length-offset` (default 0) moves them perpendicularly if needed. With `label-new-legs: #true` the new perpendicular legs (each of length a) are labelled too — their labels are pushed to the **outside** of the spiral (`new-leg-offset`, default 2.5 mm) so they never cover the drawing inside. Three modes:

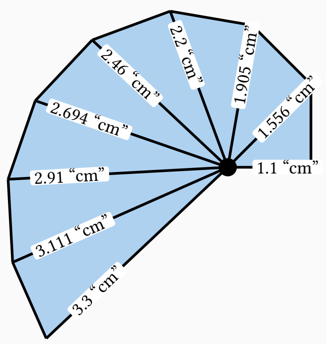
- `"values"` — decimal approximation of $a\sqrt{k}$ ("1 cm", "1.414 cm", "1.732 cm", …);
- `"formulas"` — the **exact** value $\sqrt{k}$ in units of a, simplified for perfect squares: $\sqrt{1} \to 1$, $\sqrt{4} \to 2$, …;
- a function `(k, len) => content` for full control, where `k` is the step number (1 = starting leg) and `len = sqrt(k)` in units of `size`, so the exact length is `size · len = size · sqrt(k)`.

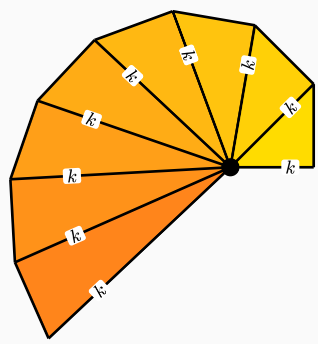




By default the labels carry a white background so they stay readable over the strokes — even when the triangles are not filled. Pass `length-label-background: #none` to remove it.

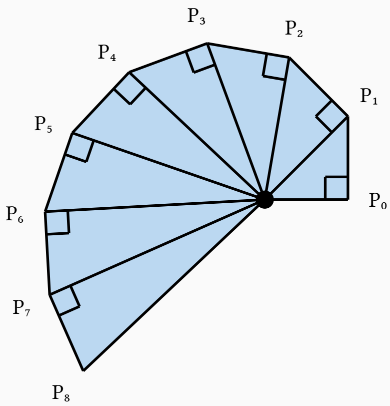
For readability on coloured fills, add a background:

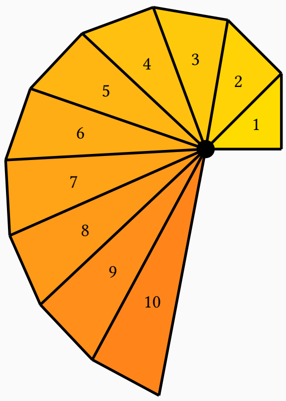




3.7. Teaching annotations

Right-angle marks, vertex labels and triangle indices:





4. Geometry helpers

- `#raw("spiral-points(steps, direction: \"ccw\", start-angle: 0deg)")` returns the vertices $P_0 \ldots P_N$ in math coordinates (y up), in units of the leg length — handy for drawing the spiral inside a CeTZ canvas or for further computations.
- `#raw("spiral-angle(steps)")` returns the accumulated angle.

[6 vertices · angle 17 = 364.7834423487936°]

5. License

MIT. This is an independent package.